Airbnb Price Prediction Report — Paris

Generated on 2026-08-25 by the automated pipeline (`make all`). Source: InsideAirbnb.

Executive summary

This report analyzes the **Paris** Airbnb market using 48,402 cleaned listings and 26 listing features. A machine-learning model was trained to predict nightly prices and then evaluated on a **holdout set** — listings it had never seen before.

The best model was **XGBoost**. On the holdout set it achieved:

- **Average error (MAE): $84** — a typical prediction is off by about this much.
- **RMSE: $153** — like MAE, but weighs large errors more heavily.
- **MAPE: 33%** — the average error as a share of the actual price.
- **R^2: 0.69** — the share of price variation the model explains (0–1).
- **55%** of predictions were within +/-25% of the true price.

How to read this report

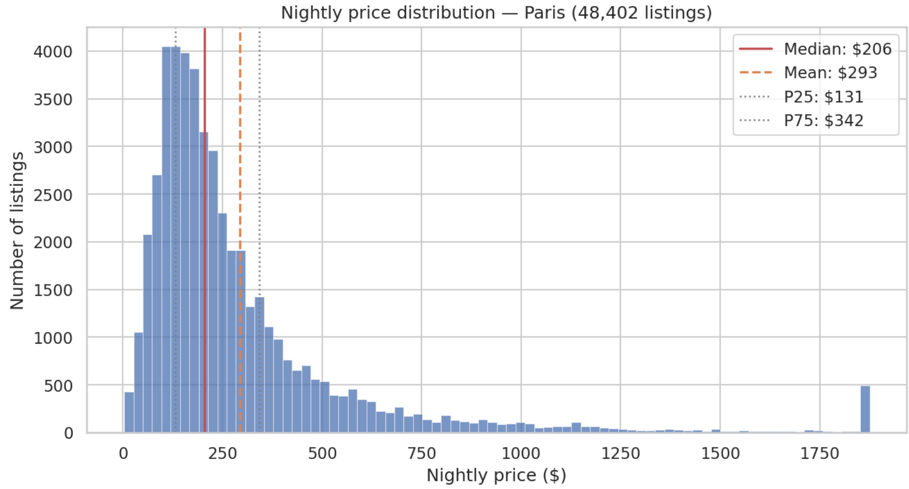
- **MAE (mean absolute error)** — the average difference between predicted and actual prices, in dollars. Lower is better; it's the easiest number to grasp ("off by about $X on average").
- **RMSE (root mean squared error)** — similar to MAE but penalizes large mistakes more. If RMSE is much higher than MAE, a few big errors are inflating it.
- **MAPE (mean absolute percentage error)** — the same idea as MAE, but expressed as a percentage of the actual price, so it can be compared across cities.
- **R^2** — how much of the variation in prices the model explains (0 = no better than guessing the average; 1 = perfect).
- **Holdout set** — the 10% of listings the model never saw while training. All accuracy numbers come from this set, so they are an honest estimate of real-world performance.
- **Top features** — the listing attributes the model leans on most when setting a price.

Data

- Source: InsideAirbnb (latest available snapshot)
- Listings after cleaning: **48,402** — median nightly price **$206**, mean **$293**
- Features used by the model: **26**
- Train / validation / holdout split: **33,881 / 8,712 / 5,809** listings (70/15/10, random seed 42)

Market overview

Price distribution



- Median nightly price: **$206**; the middle 50% of listings fall between **$131** and **$342**. - The mean (**$293**) sits well above the median — a minority of expensive listings pull the average up, so the **median is the better guide** to a typical price.

Price by room type

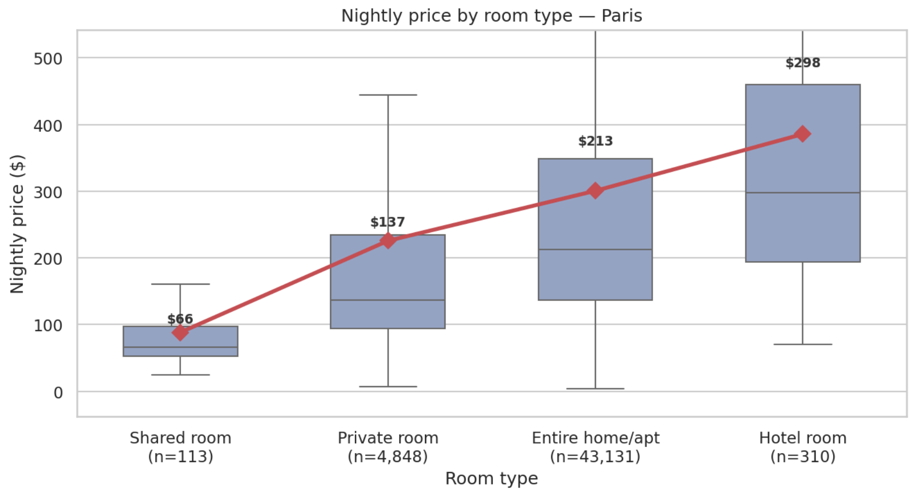


Figure 1: price_by_room_type

Room type	Listings	Median price	Mean price
Hotel room	310	$298	$386
Entire home/apt	43,131	$213	$301
Private room	4,848	$137	$226
Shared room	113	$66	$88

Price by number of bedrooms

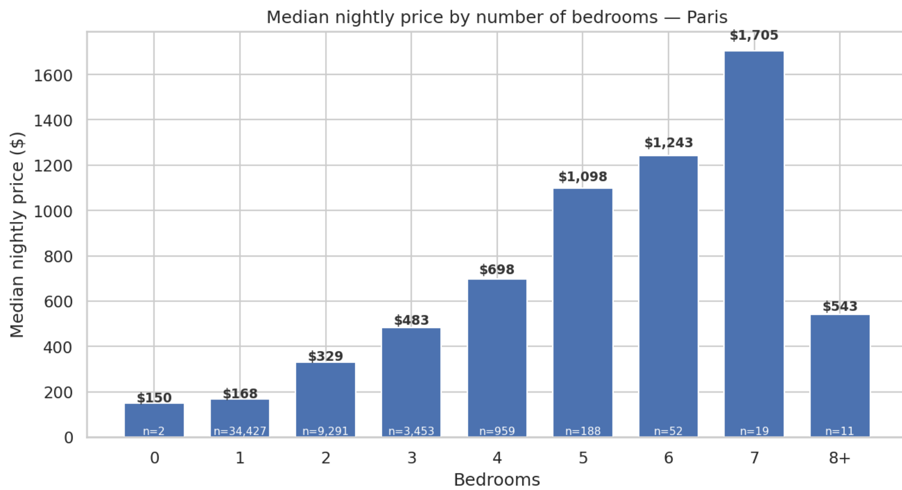


Figure 2: price_by_bedrooms

Feature correlations

Model accuracy on the holdout set

Metrics for the chosen model (**XGBoost**) on 5,809 unseen listings:

Metric	Value	What it means
MAE	$84	Average prediction is off by this much
RMSE	$153	Errors, with big misses weighted more
MAPE	33%	Average error as a % of the true price
R^2	0.69	Share of price variation explained

Accuracy in practice: 23% of predictions were within +/-10% of the true price, 55% within +/-25%, and 82% within +/-50%.

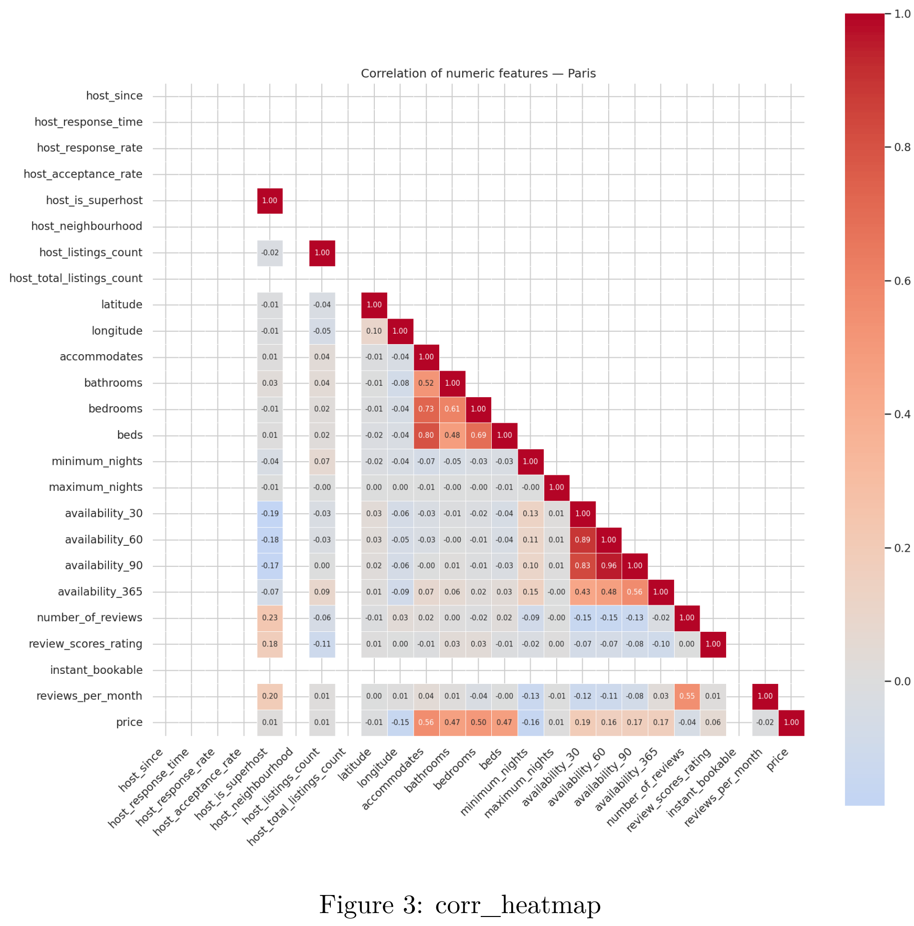
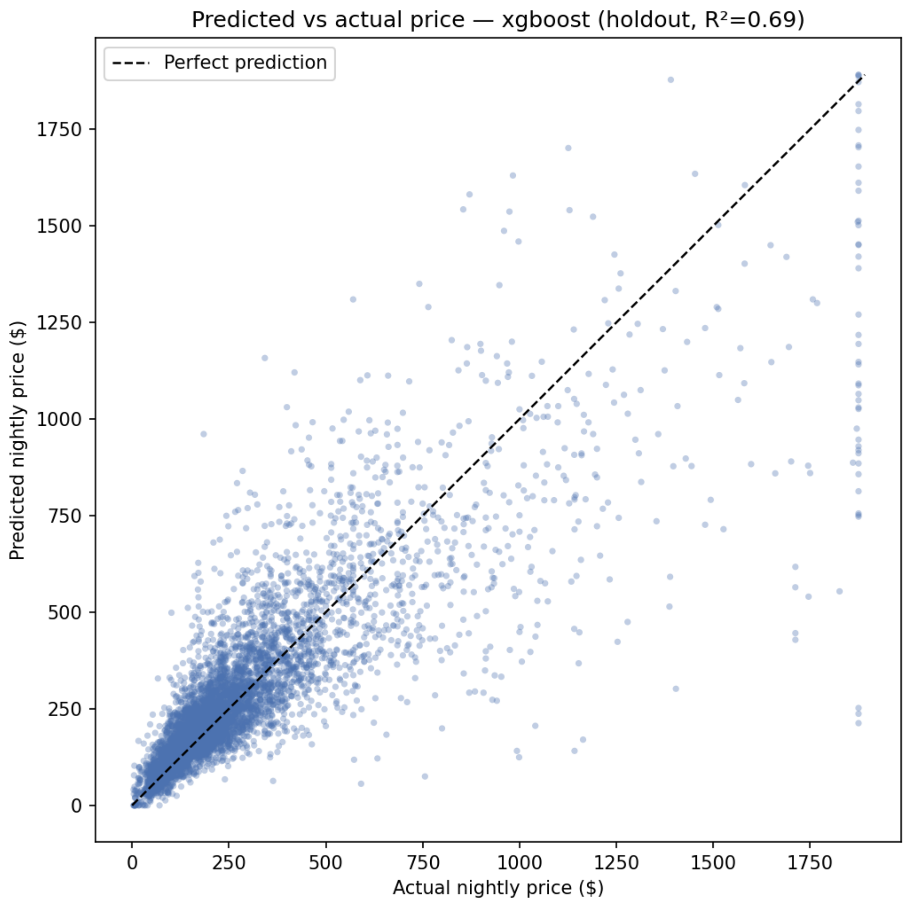
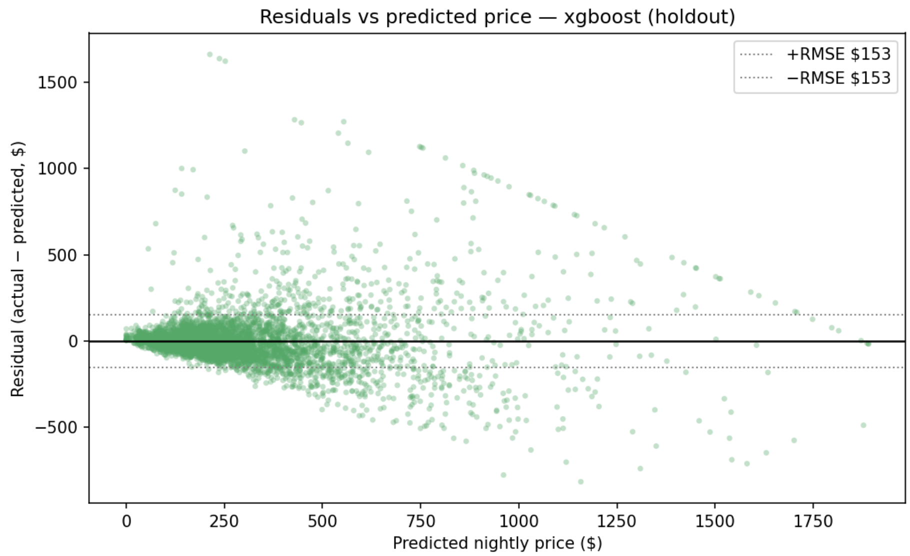


Figure 3: corr_heatmap



Points close to the dashed line are accurate predictions; points above it mean the model under-predicted.



Residual = actual - predicted. A cloud centered on the zero line means errors are balanced rather than systematically too high or too low.

Models & results

Model	Validation RMSE
Random Forest	$173
XGBoost	$158
TabPFN (capped sample, not directly comparable)	$175

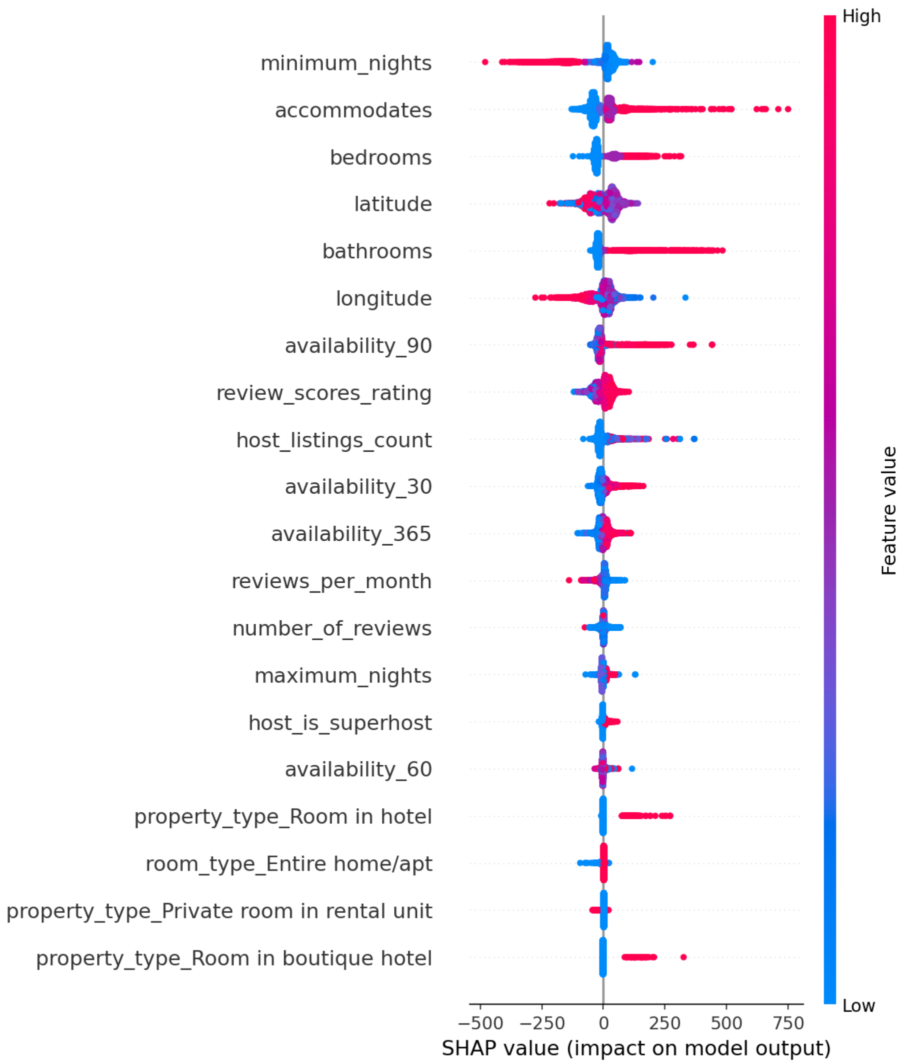
The model with the lowest validation RMSE was then scored once on the holdout set — see the accuracy section above for its final numbers.

Top features

The listing attributes that influence price the most (importance = how much the model relies on them):

Feature	Importance	Meaning
bathrooms	0.193	the number of bathrooms
accommodates	0.084	how many guests a listing sleeps
bedrooms	0.078	the number of bedrooms

Feature	Importance	Meaning
availability_90	0.061	availability over the next 90 days
minimum_nights	0.049	the minimum nights required
property_type_Room in boutique hotel	0.044	the property type (apartment, house, etc.) (specifically Room in boutique hotel)
room_type_Entire home/apt	0.043	the room type (entire home vs private/shared room) (specifically Entire home/apt)
property_type_Private room in rental unit	0.028	the property type (apartment, house, etc.) (specifically Private room in rental unit)
property_type_Room in hotel	0.026	the property type (apartment, house, etc.) (specifically Room in hotel)
room_type_Private room	0.026	the room type (entire home vs private/shared room) (specifically Private room)



SHAP summary: each dot is one listing; colour shows the feature's value (red = high, blue = low) and position shows whether it pushed the predicted price up (right) or down (left).

Prediction examples

A few real holdout listings with the model's prediction:

Listing	Actual	Predicted	Error	Error %
accommodates=2, bedrooms=1, room_type=Entire home/apt	$215	$240	$25	11.8%
accommodates=1, bedrooms=1, room_type=Entire home/apt	$47	$32	$15	32.0%
accommodates=1, bedrooms=2, room_type=Entire home/apt	$944	$1,094	$150	15.9%
accommodates=2, bedrooms=1, room_type=Entire home/apt	$110	$116	$6	5.0%
accommodates=2, bedrooms=1, room_type=Entire home/apt	$51	$79	$28	55.7%
accommodates=2, bedrooms=1, room_type=Entire home/apt	$183	$150	$32	17.6%
accommodates=2, bedrooms=1, room_type=Entire home/apt	$291	$283	$8	2.8%
accommodates=3, bedrooms=1, room_type=Entire home/apt	$500	$205	$295	58.9%

Recommendations

- The best-performing model was **XGBoost** (holdout RMSE $153, MAE $84, R^2 0.69).
- On validation, **XGBoost** beat Random Forest by 9% on RMSE ($158 vs $173). A simple ensemble or blending the two may squeeze out further gains.
- Pricing is most strongly driven by **bathrooms, accommodates, bedrooms** — i.e. bathrooms (the number of bathrooms); accommodates (how

many guests a listing sleeps); bedrooms (the number of bedrooms). Hosts can use these levers when setting nightly rates.

- On average, predictions are off by about **33%** of the actual price (median error 22%), and **55%** of predictions landed within +/-25% of the true price.
- Accuracy caveat: the holdout RMSE of $153 is **75% of the median listing price ($206)**, so predictions for typical listings are expected to be off by roughly +/-$153.
- Next steps: add richer features (neighbourhood_cleansed, amenities, review scores breakdowns), train on a log-transformed target to tame high-end outliers, run a deeper hyperparameter search, and extend the pipeline to more cities.

Limitations

- The model only "sees" the features listed in the config — it cannot capture neighbourhood charm, amenities, photos, or seasonality.
- Extreme high-end prices are clipped at the 99th percentile during cleaning, so predictions at the very top of the market are less meaningful.
- The accuracy numbers are averages: individual predictions can be much better or much worse.
- This report is generated automatically from the pipeline artifacts and reflects the data as of 2026-08-25.

Automated pipeline: fetch → preprocess → EDA → train → evaluate → report. Code in `src/`, config in `configs/base.yaml`.